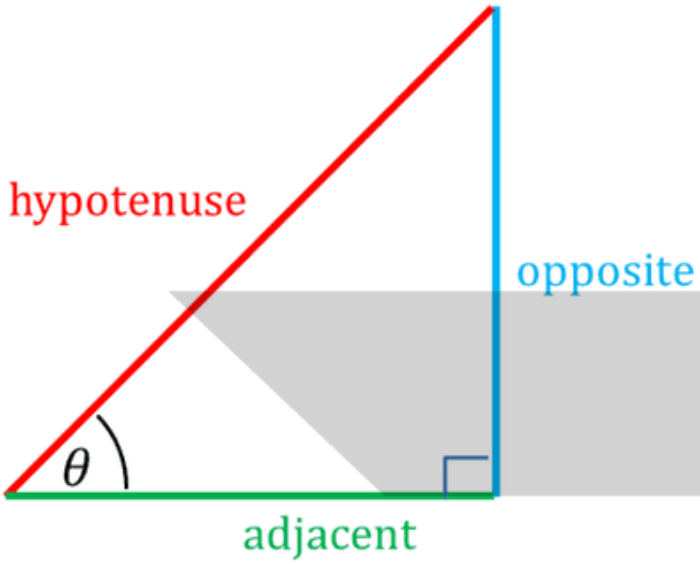
# 1. Trigonometry

## 1.1. Trigonometry ratios in right angle triangles (for θ between 0° and 90°)




$$sin(\theta) = \frac{\text{opposite}}{\text{hypotenuse}}$$

$$cos(\theta) = \frac{\text{adjacent}}{\text{hypotenuse}}$$

$$tan(\theta) = \frac{\text{opposite}}{\text{adjacent}}$$

- A mnemonic to remember the ratios is "SOH CAH TOA".

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**Example**

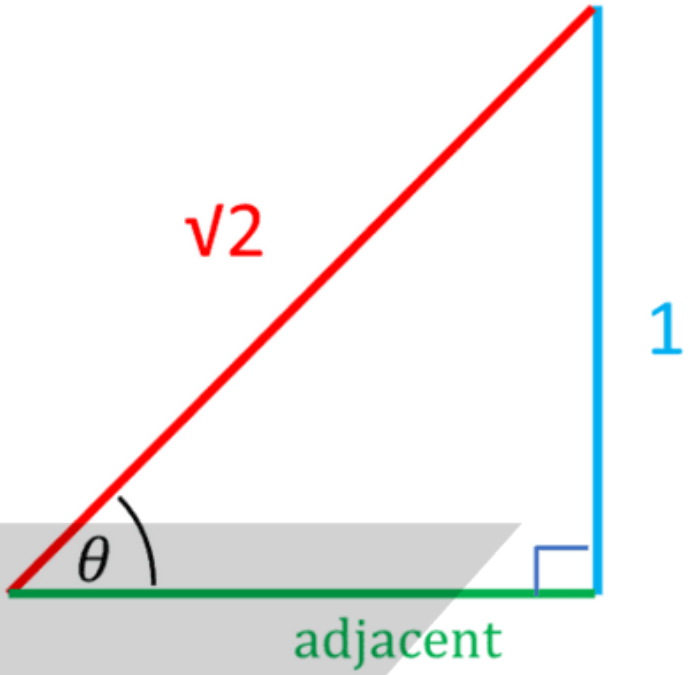
Find the ratio of $cos(\theta)$ , when $sin(\theta) = \frac{1}{\sqrt{2}}$

**Answer**

**Method 1**

- Upon comparing $sin(\theta) = \frac{1}{\sqrt{2}}$ and $sin(\theta) = \frac{\text{opposite}}{\text{hypotenuse}}$ , we note that the length of the opposite side is 1 and the length of the hypotenuse is $\sqrt{2}$ .

We can now use the Pythagoras theorem to find the adjacent side of the triangle:




$$(\text{adjacent})^2 + (1)^2 = (\sqrt{2})^2 = 1$$

$$cos(\theta) = \frac{\text{adjacent}}{\text{hypotenuse}} = \frac{1}{\sqrt{2}}$$

**Method 2**

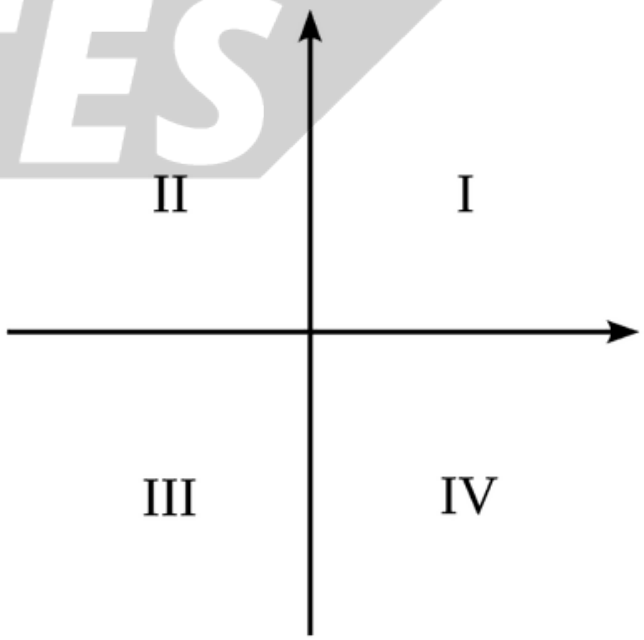
Using the fact that

$$sin(\theta) = \frac{1}{\sqrt{2}} \implies \theta = sin^{-1}\left(\frac{1}{\sqrt{2}}\right) = 45^\circ$$

$$cos(45^\circ) = \frac{1}{\sqrt{2}} \implies cos(\theta) = \frac{1}{\sqrt{2}}$$

## 1.2. Quadrants

- The Cartesian plane is divided into 4 quadrants \n



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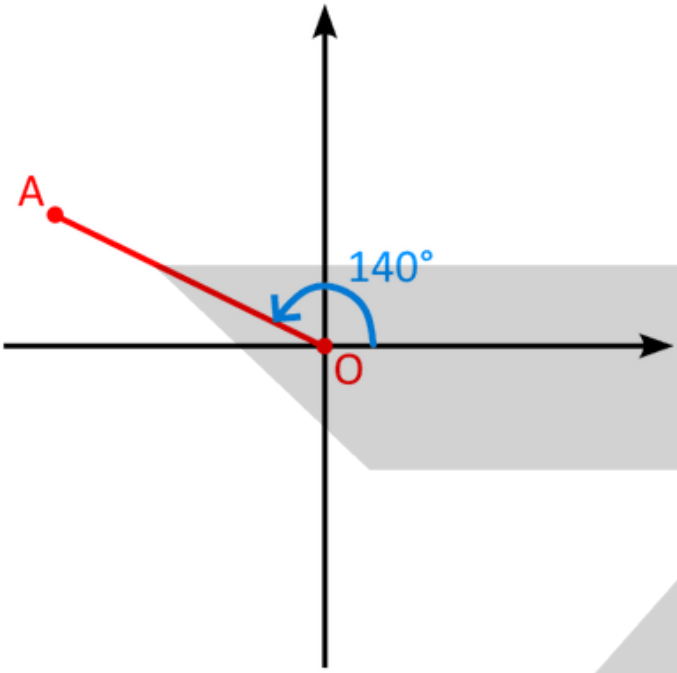
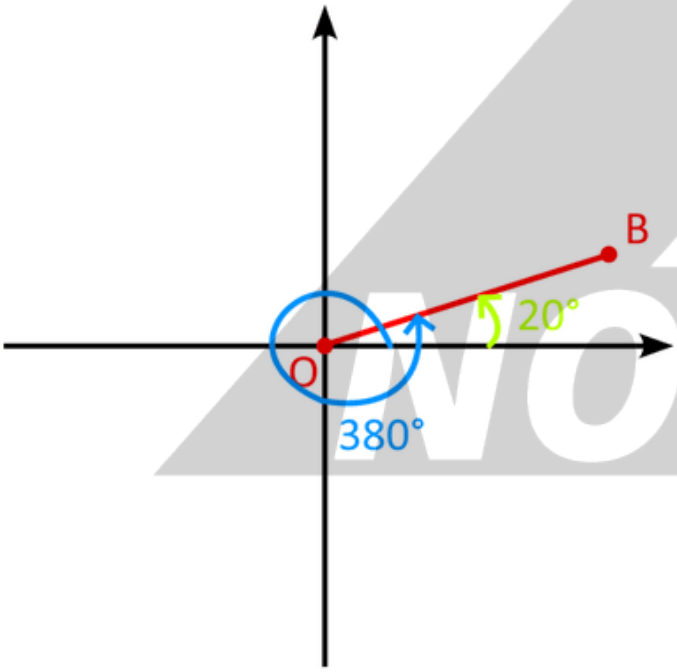
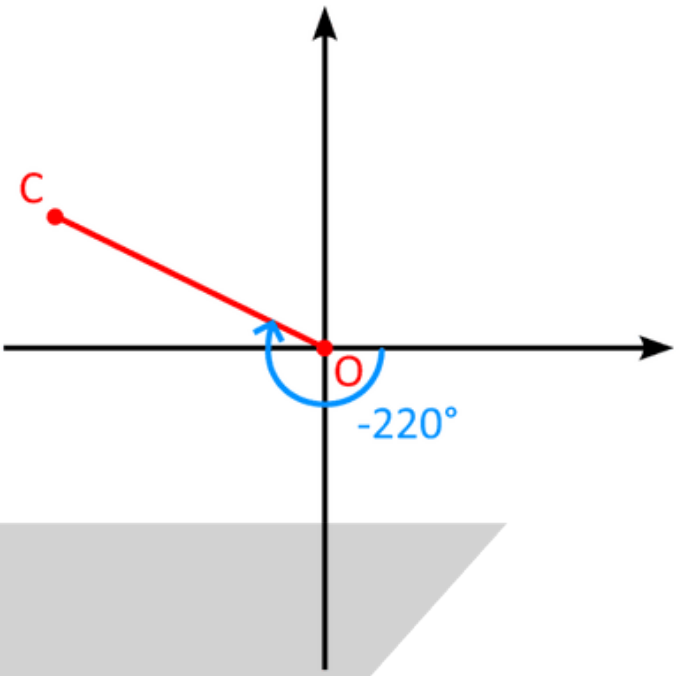
- an anticlockwise rotation from the x-axis of the first quadrant shows an increase in angle, but clockwise rotation shows a decrease in angle

**Example**

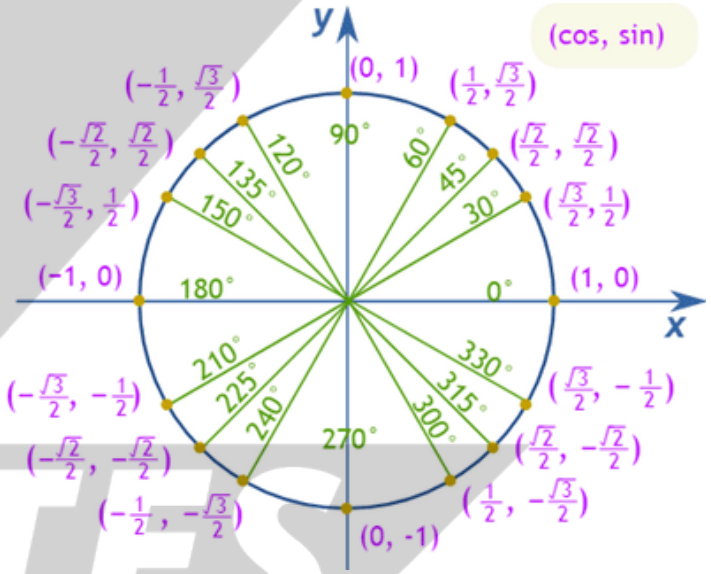
draw a line on a graph OA which is rotated 140°, a line OB rotated 380°, and a line OC rotated -220° Answer:




## 1.3. Quadrant - continued




- First quadrant
  - $cos(\theta) = +ve$
  - $sin(\theta) = +ve$
  - $tan(\theta) = +ve$
- Second quadrant
  - $cos(\theta) = -ve$
  - $sin(\theta) = +ve$
  - $tan(\theta) = -ve$
- Third quadrant
  - $cos(\theta) = -ve$
  - $sin(\theta) = -ve$
  - $tan(\theta) = +ve$



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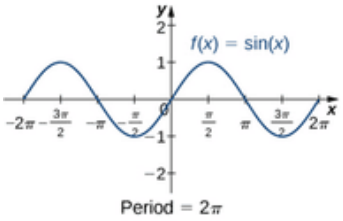
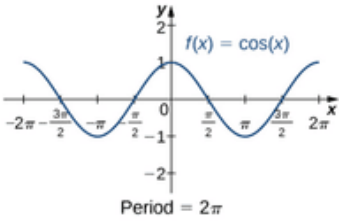
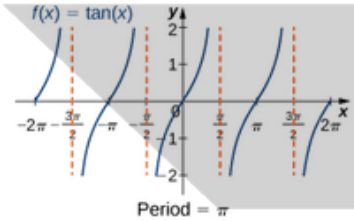
- Fourth quadrant
  - $cos(\theta) = +ve$
  - $sin(\theta) = -ve$
  - $tan(\theta) = -ve$

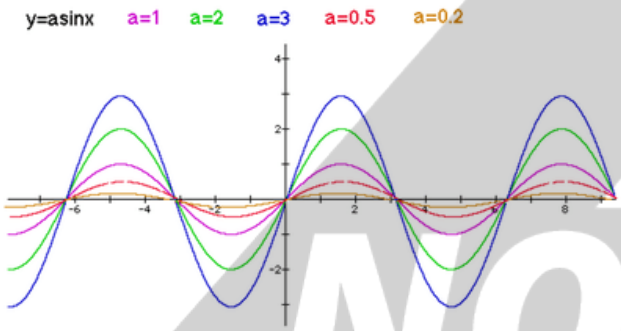
## 1.4. Graphs of Trigonometric functions



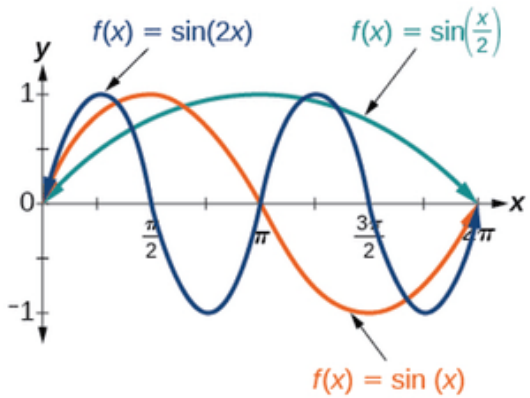



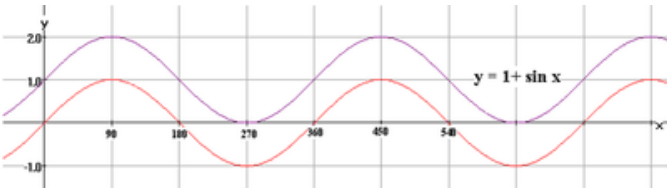

- period is the length of one cycle of the graph
- amplitude is the distance between the maximum or minimum points from the principal axis
- graph of $y = asin(x)$
  - The graph is stretched along the y-axis with a factor of $a$
  - The range of the graph changes to $a < y < -a$



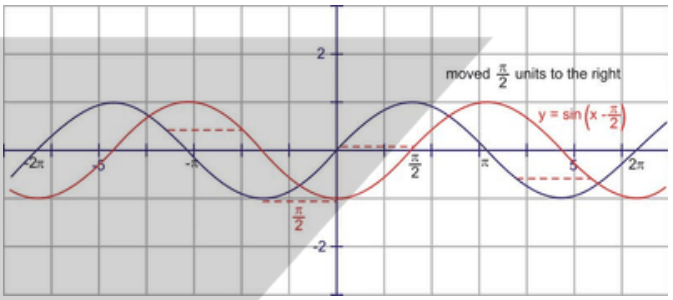

- graph of $y = sin(ax)$
  - the period of the graph is $\frac{2\pi}{a}$
  - the graph is stretched along the x-axis by a factor of $\frac{1}{a}$




- graph of $y = sin(x) + a$
  - the graph is shifted along the y-axis by the value of $a$
  - translation of the graph by [0, $a$]
  - The range of the graph changes to $-1 + a < y < 1 + a$




- graph of $y = sin(x + a)$
  - the graph is shifted along the x-axis by the value of -$a$
  - translation of the graph by [-$a$, 0]




- When working with multiple transformations, go with this order:
  - firstly, work with the transformation inside the brackets of the trigonometric function
  - secondly, apply the transformation multiplied/divided to the trigonometric function
  - lastly, apply the transformation which is being added/subtracted to the trigonometric function.

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### Example

Draw the graph of $y = 3cos(x - \frac{1}{2}\pi) - 1$

### Answer

- draw the graph of $y = cos(x)$



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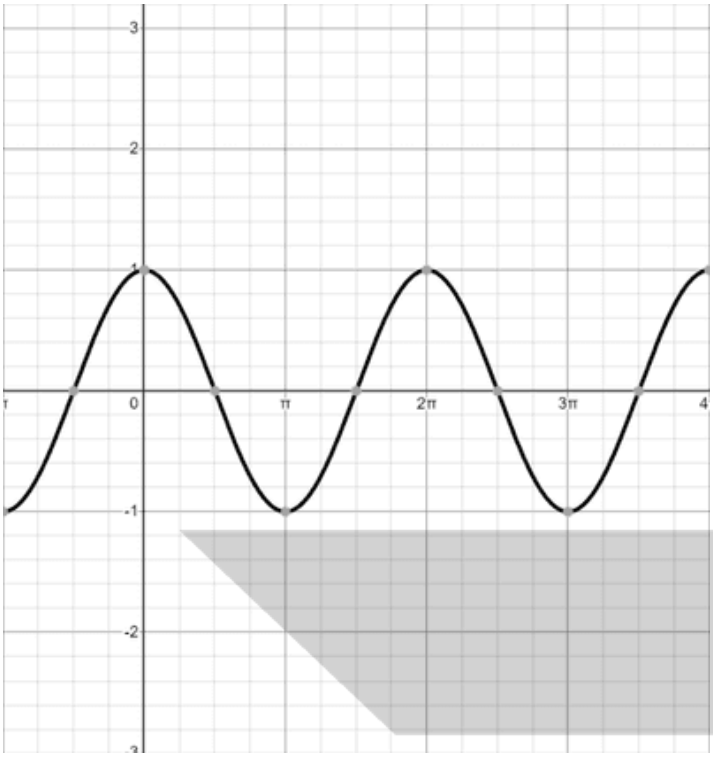
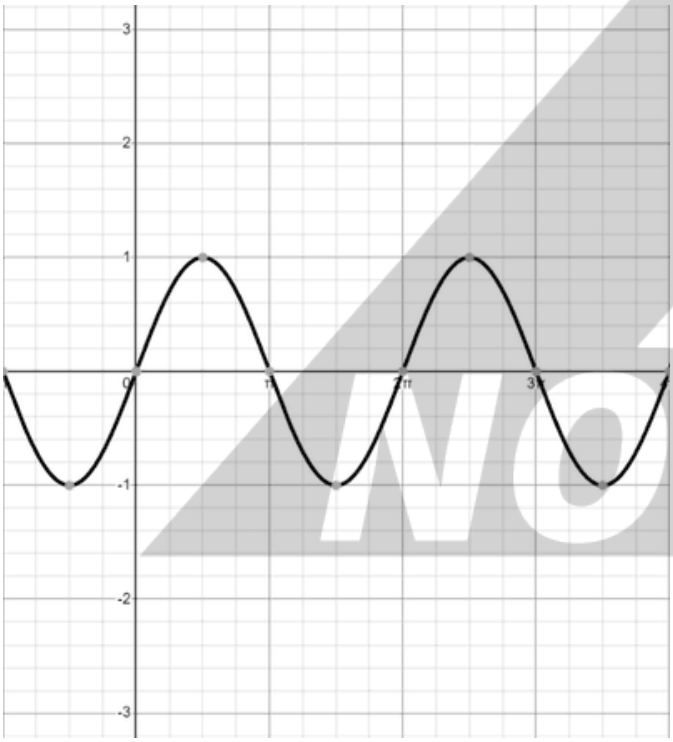
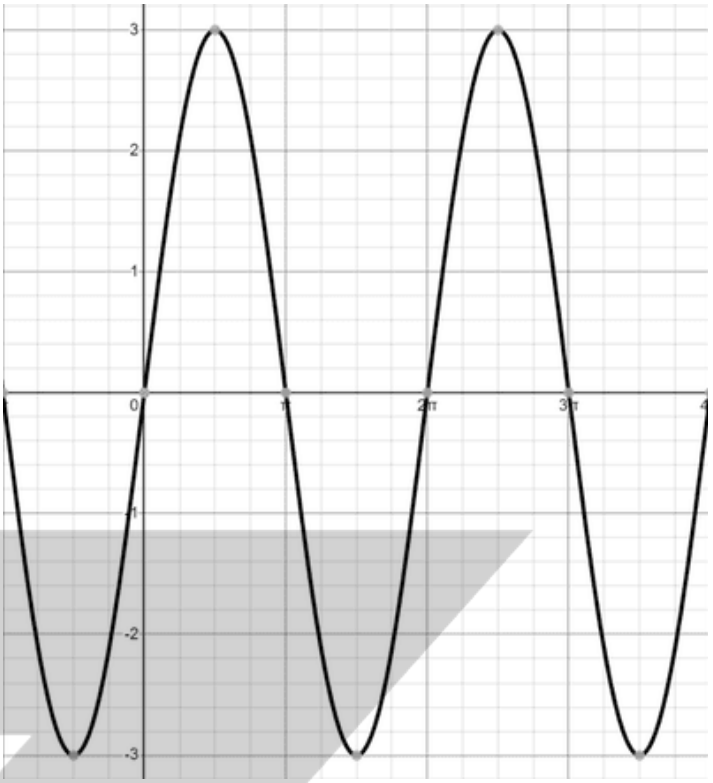



- apply the transformation inside the brackets $(x-90)$ , which translates the graph by

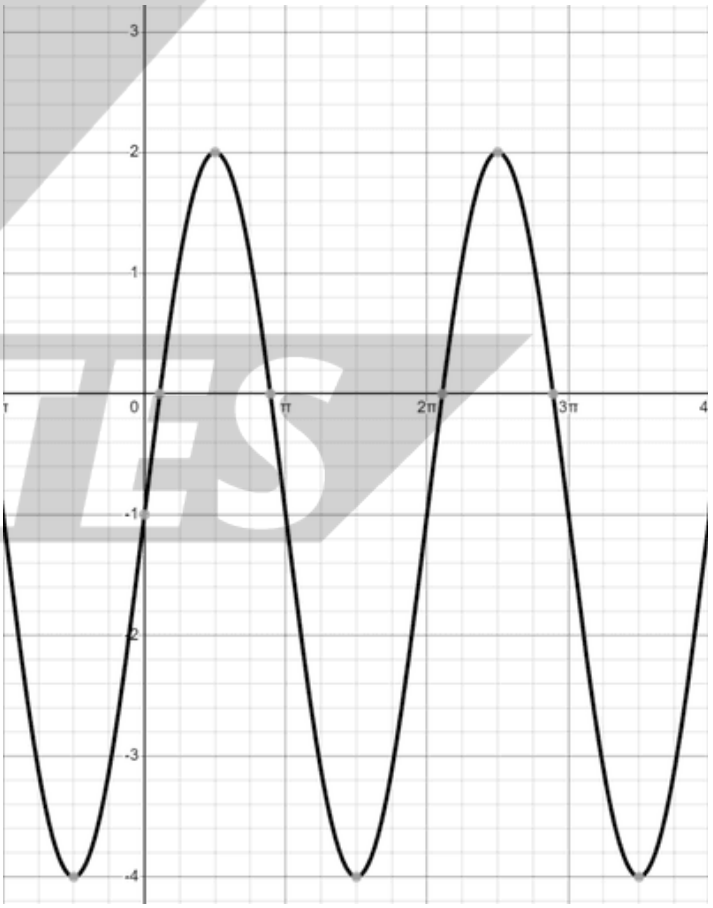
$(90, 0)$




- apply the transformation which is being multiplied/divided to the trig function, in this case 3, which should stretch the graph by a factor of 3 along the y-axis




- finally, apply the transformation that is being added/subtracted to the expression, in this case it is $-1$, which would translate the graph by $(0, -1)$



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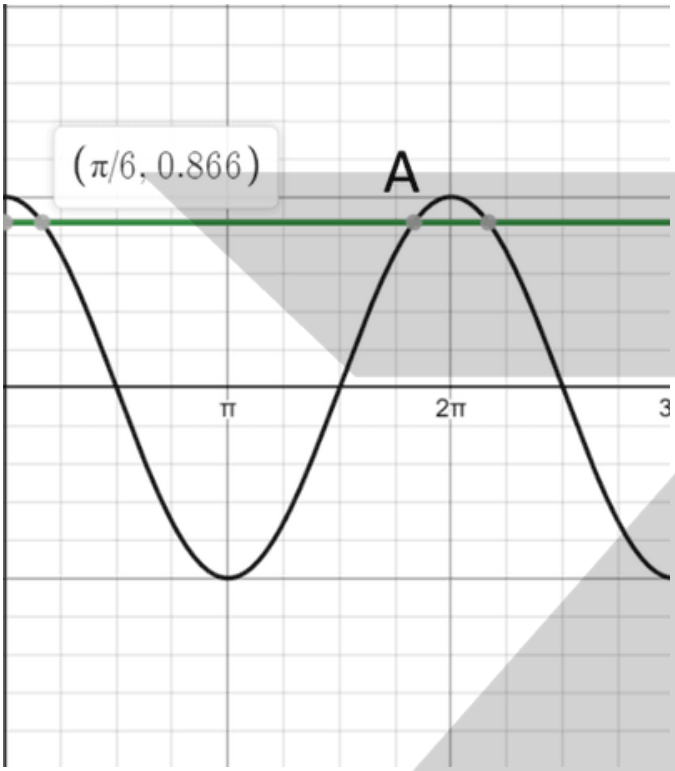
### 1.5. Find the other angles from the principle angle

- in case of $cos(x) = \frac{\sqrt{3}}{2}$ , we know using a calculator or by memory that $x = 30^\circ$ or $x = \frac{\pi}{6}$ , but this is only the principle angle and there are many more solutions to this.
- to find the other values, draw a graph and draw a horizontal line for the value given to find the angles at which the trigonometric function give the value stated. in the previous example, the graph would look like this




- as you can see there are also other points on the graph which intersect with the horizontal line, these are also additional solutions
- in the question, a range might be given and you would have to find all the solutions in that given range.
- in this case, to find the other point $A$ we can do $2\pi - \frac{\pi}{6}$ , which equals to $\frac{11\pi}{6}$
- here we are using the property of trigonometry functions that they are symmetrical, and thus such calculations can be used to find the angles

### 1.6. Trigonometry identities

$$tan(\theta) = \frac{sin(\theta)}{cos(\theta)}$$

$$sin^2(\theta) + cos^2(\theta) = 1$$

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Example

Prove the identity $\frac{cos^2(x) - sin^2(x)}{cos(x)} + \frac{1}{cos(x)} \equiv 2cos(x)$

Answer

$sin^2(x)$ can be written as $1 - cos^2(x)$

$$\frac{cos^2(x) - (1 - cos^2(x))}{cos(x)} + \frac{1}{cos(x)}$$

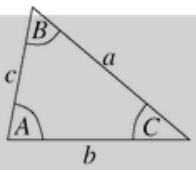
$$\frac{2cos^2(x) - 1}{cos(x)} + \frac{1}{cos(x)}$$

$$\frac{2cos^2(x) - 1 + 1}{cos(x)}$$

$$\frac{2cos^2(x)}{cos(x)} \implies 2cos(x)$$

### 1.7. Trigonometry with non right-angled triangles

- in these cases, you can either use the sine rule or the cosine rule



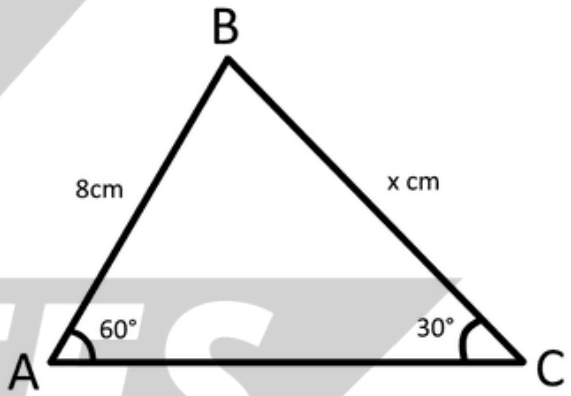

Sine Rule

$\frac{a}{\sin(A)} = \frac{b}{\sin(B)} = \frac{c}{\sin(C)}$ (for finding sides) or $\frac{\sin(A)}{a} = \frac{\sin(B)}{b} = \frac{\sin(C)}{c}$ (for finding angles)

Cosine Rule

$a^2 = b^2 + c^2 - 2bc\cos(A)$ (for finding sides) or $\cos(A) = \frac{b^2 + c^2 - a^2}{2bc}$ (for finding angles)

Example




Find the value of $x$

Answer

in this question, we are given two angles and one of the sides, where we have to find the length of the other side.

So we know we should use the sine rule $\frac{a}{sin(A)} = \frac{b}{sin(B)} = \frac{c}{sin(C)}$

we can write this as $\frac{8}{sin(30)} = \frac{x}{sin(60)}$

cross multiplying it would give us

$8sin(60) = xsin(30)$

$8 * \frac{\sqrt{3}}{2} = \frac{1}{2}x$

$8\sqrt{3} = x$



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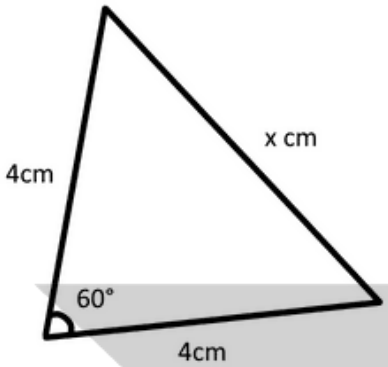
$\therefore\ x = 8\sqrt{3}$

**Example**




Find the value of $x$

Answer:

in this question, we are given two side lengths and the angle between them

in this case, we shall use the cosine rule

$a^2 = b^2 + c^2 - (2bc * cos(A))$

we can write this as $x^2 = 4^2 + 4^2 - (2 * 4 * 4 * cos(60))$

$x^2 = 16 + 16 - (16)$

$x^2 = 16$

$\therefore\ x = 4$



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